IN CLOSING

Ourselves

After having journeyed so far, from the structure of deep space to the margins of the known cosmos, I would like to return, before closing this series of lessons, to the subject of ourselves.

What role do we have as human beings who perceive, make decisions, laugh and cry, in this great fresco of the world as depicted by contemporary physics? If the world is a swarm of ephemeral quanta of space and matter, a great jigsaw puzzle of space and elementary particles, then what are we? Do we also consist only of quanta and particles? If so, then from where do we get that sense of individual existence and unique selfhood to which we can all testify? And what then are our values, our dreams, our emotions, our individual knowledge? What are we, in this boundless and glowing world?

I cannot even imagine attempting to really answer such a question in these simple pages. It's a tough question. In the big picture of contemporary science there are many things that we do not understand, and one of the things which we understand least about is ourselves. But to avoid this question or to ignore it would be, I think, to overlook something essential. I've set out to show how the world looks in the light of science, and we are a part of that world too.

'We', human beings, are first and foremost the subjects who do the observing of this world; the collective makers of the photograph of reality which I have tried to compose. We are nodes in a network of exchanges (of which this present book is an example) through which we pass images, tools, information and knowledge.

But we are also an integral part of the world which we perceive; we are not external observers. We are situated within it. Our view of it is from within its midst. We are made up of the same atoms and the same light signals as are exchanged between pine trees in the mountains and stars in the galaxies.

As our knowledge has grown we have learnt that our being is only a part of the universe, and a small part at that.

This has been increasingly apparent for centuries, but especially so during the last century. We believed that we were on a planet at the centre of the